

Optical Fibres

Student Notes

1 Experimental Aim

The aim of this project is to study the fundamental properties of single-mode and multi-mode fibres. You will practice cleaving fibre ends and will couple laser light into multi-mode and single-mode fibres. Basic fibre parameters, such as numerical aperture, attenuation coefficient and far-field power distribution will be measured.

2 Introduction

Optical fibres are circular dielectric waveguides which can transport optical energy and information. The wide range of applications of optical fibres stems from their characteristic properties such as light weight, low loss, immunity to electromagnetic interference, high security against intrusion and a large information carrying capacity.

3 Theory

3.1 Step and Graded Index Fibres

The difference between step and graded index fibres is the refractive index of the core of the fibre. A step index fibre has a uniform refractive index through out the core and undergoes an abrupt (step) change at the core-cladding boundary. In a graded index fibre the core is made to vary gradually such that it is a maximum at the centre of the core.

The graded core allows for less dispersion due to the light rays travelling at different velocities in different paths because of the refractive index variation. This allows the light rays to reach the end of the fibre at approximately the same time and thus there is no dispersion. The variations in path length through the step index fibres cause dispersion, and hence a distorted output.

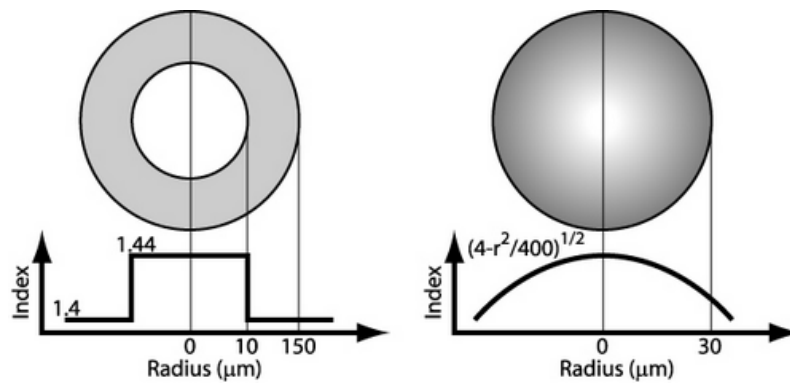


Figure 1: Refractive index of step index fibre (left) and graded index fibre (right).

3.2 Single and Multimode Fibres

Single mode fibres have a small core size that permits only one mode or ray of light to be transmitted. The small size of the fibre core causes the light to only reflect a little as it passes throughout and hence there is a lower amount of attenuation and the signal can travel further.

Multimode fibres have larger cores that guide many modes simultaneously and so more data, in the form of light signals, can pass through at any one time. The larger core diameter creates more light reflections and higher dispersion and attenuation rates limiting the quality of the signal over long distances.

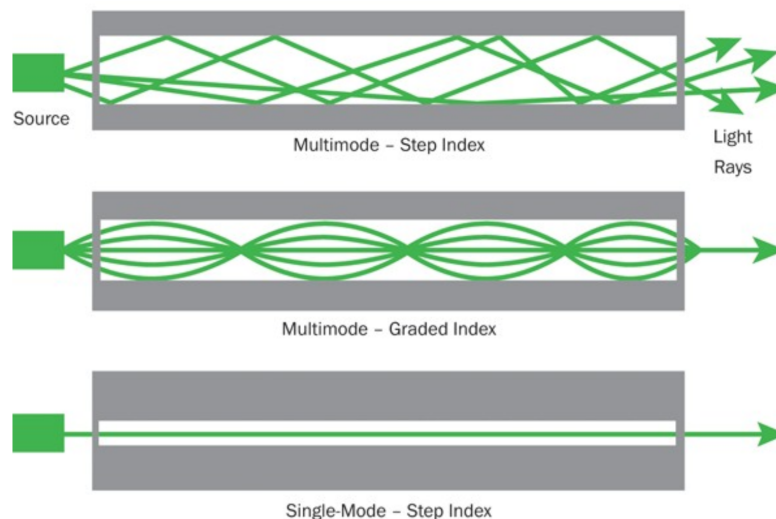


Figure 2: Multimode and single-mode light propagation.

4 Experiment 1: Preparation of Fibre Ends

A clean, mirror-flat and perpendicular end-face is required from most fibre applications. One method of obtaining this is cleaving, a deliberate controlled break. A good cleave is required for a successful low loss coupling of a fibre. In this experiment a F-MLD multimode fibre and a F-SV single-mode fibre were prepared by cleaving for the later experiments. The method of preparation and comments on this process are laid out below.

4.1 Experimental Fibre Parameters

Two types of Newport optical fibre will be used in these experiments: F-MLD (graded-index, multimode) and F-SV (step-index, single-mode).

Table 1: Fibre Parameters

Model No.	Operating Wavelength (nm)	Core Diameter (μm)	Cladding Diameter (μm)	Jacket Diameter (μm)	NA	Attenuation (dB/km)
F-MLD	850-1300	100	140	500	0.3	4.5-2.0
F-SV	633	4	125	250	0.1	<12

4.2 Experimental Instructions

In this experiment you will cleave the ends of F-MLD and F-SV fibre segments to be used throughout the other experiments. The F-MLD fibre will be used in experiment 2, and the F-SV fibre will be used in experiment 4.

For specific method please see *Operating Instructions*.

5 Experiment 2: Measurement of Numerical Aperture

For an optical fibre there exists a maximum angle of incidence beyond which, light does not propagate through the fibre but rather is refracted in to the cladding medium. This angle, θ , is known as the maximum angle of acceptance and $n_0 \sin \theta$ is termed the numerical aperture (NA), where n_0 is the outside refractive index (e.g. the air, not the core or cladding).

Qualitatively, NA is a measure of the gathering capacity of an optical fibre. For a step-index fibre and for air as the incident medium, the numerical aperture is related to the refractive indices of the core (n_{core}) and cladding (n_{cl}) by the relation:

$$NA = \frac{1}{\sqrt{n_{core}^2 - n_{cl}^2}} \quad (1)$$

$$= \frac{n_{core}}{2\Delta} \quad (2)$$

Where $\Delta = \frac{(n_{core} - n_{cl})}{n_{core}}$ is the fractional index difference.

For silica based fibres, values of NA range from about 0.1 for single-mode fibres to 0.2-0.3 for multimode communication fibres up to about 0.5 for large core fibres.

5.1 Experimental Instructions

In this experiment you will determine, experimentally, the numerical aperture of the F-MLD fibre. The numerical aperture will be measure by two methods:

- a) Using the relationship between the incident angle and power accepted by the fibre
- b) Using the far-field distribution of the fibre.

For part a), the light from the laser can be considered as a plane wave propagating in the z-direction as the width of the laser beam is much larger than the diameter of the fibre core.

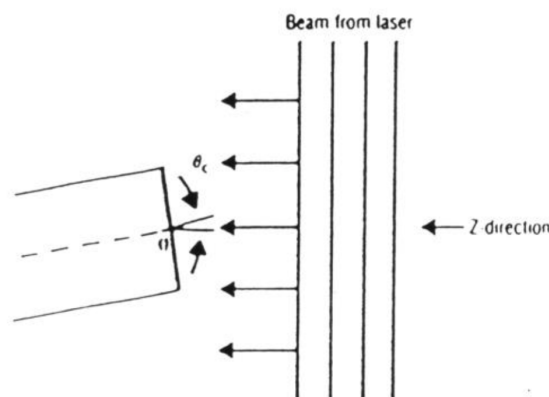


Figure 3: Geometry of a plane-wave launch of a laser beam into an optical fibre.

If the fibre end face is then rotated about the point O in Figure 3, you can then measure the amount of light accepted by the fibre as a function of the incident angle θ . The Electronic Industries Association uses the sine of the angle at which the accepted power has fallen to 5% of the peak accepted power as the definition of the experimentally determined NA.

For part b), you will measure the spot size (W) of the fibre output and a distance (L) from the fibre end to approximate the NA.

For specific method please see *Operating Instructions*.

5.2 Question 1

An increase in the numerical aperture of a fibre can be achieved by increasing the refractive index of the core with respect to the cladding. What are the drawbacks in doing so?

5.3 Question 2

Explain the terms modal dispersion and bandwidth of a fibre. Why do graded-index fibres have a larger bandwidth than step-index fibres?

6 Experiment 3: Optical Fibre Attenuation

Optical power propagating in a fibre decreases exponentially with distance. The attenuation of an optical fibre is the amount of light lost between its input and output ends. The amount of optical power transmitted through a length, z (km), of a fibre is given by:

$$P_f = P_i 10^{-\frac{\Gamma z}{10}} \quad (3)$$

Where:

P_f = final power

P_i = initial power

z = distance from the origin

Γ = the attenuation per unit length, measured in decibels per kilometre (dB/km)

Attenuation (Γ) is one of the most important fibre parameters. Equation (3) can be written in terms of Γ as:

$$\Gamma = \frac{-10}{z} \left[\log \left(\frac{P_f}{P_i} \right) \right] \quad (4)$$

The loss, L , in decibels (dB) can be expressed as:

$$L = -10 \log \left(\frac{P_f}{P_i} \right) \quad (5)$$

The minus sign causes the loss to be expressed as a positive number.

6.1 Experimental Instructions

The method which is generally used to measure Γ is called the "cutback" method. The basic principle is to launch power from a source into a long length of fibre, and the power output at the far end is measured as the output power, P_f . Then, after cutting a shorter length from the input end, the power transmitted through that shorter length is measured as the input power, P_i . The schematic of laboratory set-up for this method is shown in Figure 4.

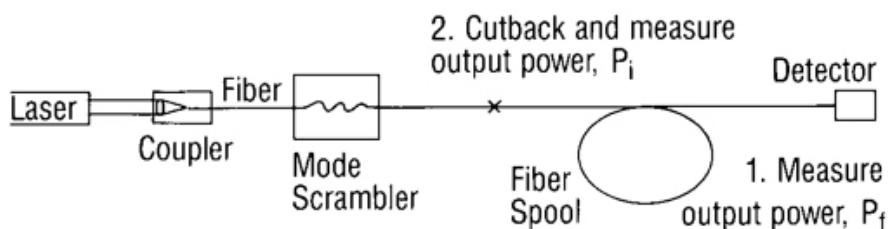


Figure 4: Schematic of cutback method for determining fibre attenuation.

In this experiment you will be performing the cutback method both with and without the use of a mode scrambler. The measurements of output power will be taken in the following order to ensure launch conditions and mode scrambling remain consistent;

1. long length of fibre, no mode scrambling
2. long length of fibre, with mode scrambling
3. short length of fibre, with mode scrambling
4. short length of fibre, no mode scrambling

For specific method please see *Operating Instructions*.

6.2 Question 3

The cutback method is not suitable for low-loss fibres. Why?

6.3 Question 4

In launching light into a fibre, what is meant by underfilled and overfilled conditions?

6.4 Question 5

Compare the attenuation coefficients of a fibre measured in the underfilled condition and that of the same fibre with a stable distribution?

6.5 Question 6

If the fibre coating (jacket) has a smaller refractive index than the cladding, how can you strip off the cladding modes?

7 Experiment 4: Single-Mode Fibres

An important quantity which determines the modes of an electromagnetic field supported by a fibre is known as V-number and is given by:

$$V = k_f a \text{NA} \quad (6)$$

Where:

$k_f = \frac{2\pi}{\lambda}$ = free space wave number

a = the radius of the core

NA = numerical aperture

When $V < 2.405$ only a single mode, the HE_{11} mode, will propagate through the fibre. The Wavelength at which V is equal to 2.405 is called the cut-off wavelength (i.e. the wavelength at which the next higher-order mode is cut off).

The lowest order mode (HE_{11}) in a fibre has a **Gaussian irradiance pattern** if the diameter of the core is large compared to the wavelength of light. That is, the irradiance as a function of distance from the beam has the form:

$$I(r) = I(0)e^{(-\frac{r}{r_0})^2} \quad (7)$$

Where:

$I(0)$ = the irradiance at the centre of the beam

r_0 = the radius at which the irradiance is $\frac{1}{e^2}$ of $I(0)$.

It is a measure of the width of the beam.

Coupling light into single-mode fibre is relatively difficult compared to multimode fibres. In addition to the precise alignment of the fibre to the laser beam, it is also necessary to match the spot size of the laser beam to the Gaussian spot size of the HE_{11} mode which will propagate through the fibre. The $\frac{1}{e^2}$ spatial half-width of the HE_{11} mode is given by:

$$w_0 = a(0.65 + 1.619V^{-1.5} + 2.879V^{-6}) \quad (8)$$

Where a is the fibre core radius.

When coupling light into a fibre, a microscope objective is commonly used to focus the collimated laser beam to a fine spot. The diameter, d_1 , of the spot size at the waist of the focused laser beam is given by:

$$d_1 = \frac{4\lambda f}{\pi d} \quad (9)$$

Where:

f = the focal length of the microscope objective lens

d = the diameter of the laser beam at the aperture of the objective lens

d is related to the divergence, θ , of the laser beam, the diameter, d_o , of the laser beam at the output of the laser, and the laser-to-lens distance, z , through the relation:

$$d = d_o \sqrt{1 + \left(\frac{z\theta}{d_o}\right)^2} \quad (10)$$

Use equations (6) - (10) to answer questions 7 - 9

7.1 Question 7

Taking the core diameter as $4\text{ }\mu\text{m}$ and NA as 0.11 for the Newport F-SV fibre, show that this fibre can accommodate only a single mode at 633 nm.

7.2 Question 8

Calculate the Gaussian spot size (w_o) of the HE_{11} mode in an F-SV fibre.

7.3 Question 9

Given that $\theta = 1.3\text{ mrad}$, $d_o = 0.7\text{ mm}$ and $f = 8\text{ mm}$, estimate the distance, z , between the laser and the objective lens to optimise the coupling efficiency of 633 nm HeNe laser light into the F-SV fibre.

7.4 Experimental Instructions

In this experiment you will use the gaussian approximation of the far-field distribution of the F-SV single-mode fibre to find its numerical aperture experimentally. You will measure and plot the intensity of the output light as a function of angle.

For specific method please see *Operating Instructions*.

7.5 Question 10

Why is the output end rotated instead of the launch end for single-mode fibres?